

And *nestedLoopPermutations.html* contains the equivalent JavaScript program:

JavaScript

```
<script>
for (a of ['A', 'B', 'C', 'D', 'E']) {
  for (b of ['A', 'B', 'C', 'D', 'E']) {
    for (c of ['A', 'B', 'C', 'D', 'E']) {
      for (d of ['A', 'B', 'C', 'D', 'E']) {
        document.write(a + b + c + d + "<br />")
      }
    }
  }
}
</script>
```

The output of these programs looks like this:

```
A A A A
A A A B
A A A C
A A A D
A A A E
A A B A
A A B B
--snip--
E E E C
E E E D
E E E E
```

The problem with generating permutations with four nested loops is that it works only for permutations that are exactly four characters. Nested loops cannot generate permutations for arbitrary lengths. Instead, we can use a recursive function, as described in the next section.

You can remember the difference between permutations with and without repetition with the examples in this chapter. Permutations *without* repetition go through all possible orderings of the elements in a set, like our wedding guest seating chart example. Permutations *with* repetition go through all the possible combinations of a combination lock; the order matters, and the same element can appear more than once.